

GO WITH THE FLOW: OPTIMAL SAILBOAT RACING

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ABSTRACT. In this paper we will try to model the optimal path in the historical M&M 100 Miler sailing race. Many forces impact optimal sailing, including wind direction, current, and waves, making this a challenging problem to model accurately. The primary questions we work to answer are what the simplest model is that rediscovers real sailing techniques, such as tacking, and how many parameters we can add into our model while still being able to solve it and get reasonable results. We solve the classic Zermelo Navigation Problem assuming constant boat velocity and fixed current drift vector field. Further iterations involve relying on wind instead of a motor, taking into account the impact of the islands on wind and current speed, and adding land penalties. Our more accurate models were able to find a path that matched the speeds of real-world sailors from the year we got current and wind data.

1. BACKGROUND

The historic M&M 100 Miler sailing race takes place annually in Lake Michigan.

The primary question this paper seeks to determine is what the simplest model is that re-discovers real world sailing techniques, particularly tacking and jibing.

2. MATHEMATICAL REPRESENTATION

In this section we go over the models we used to represent sailing.

Model 1: Zermelo Navigation Approach.

Model 2: Square-Sail Dynamics.

Model 3: Triangular-Sail Dynamics.

Model 4: Sailing With Land Penalties.

3. SOLUTION

[CM26]

4. INTERPRETATION

Appropriateness of models.

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Potential improvements.

Impact.

5. CONCLUSION

While the simplest two models we worked with were not able to rediscover real sailing techniques, the more complicated ones were.

6. FERPA RELEASE

We give Dr. Jarvis, and other instructors at BYU teaching ACME classes permission to share our project as an example of a good project in future classes they teach.

- Ethan Palenske

REFERENCES

- [CM26] Jackson Pond Connor McBride, Ethan Palenske. vol4-second-project. <https://github.com/EthanPrimes/vol4-second-project>, 2026. GitHub repository, MIT license, accessed 2026-4-11.